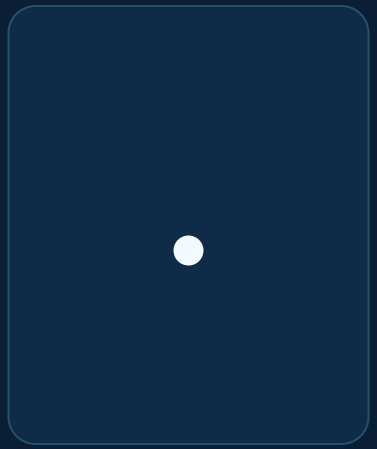
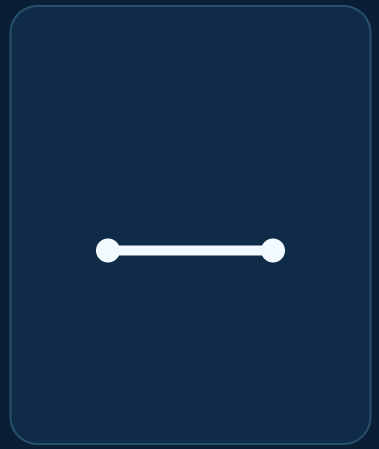
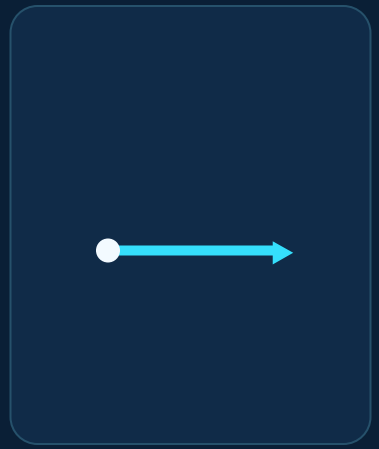
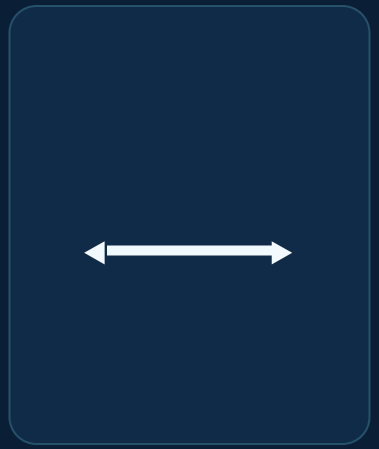
1. Geometry starts with a few simple objects.
2. A point marks an exact position.
3. Two endpoints make a line segment.
4. A ray starts at one point and goes on in one direction.
5. A line goes on in both directions.
6. Match each name to its drawing.
7. *Feedback:* A point has no length — it only marks a position.
8. *Feedback:* A line segment stops at two endpoints.
9. *Feedback:* A ray has exactly one endpoint and one arrow.
10. *Feedback:* A line has no endpoints — it goes on both ways.
11. All matched — endpoints and arrows tell them apart.

Match each name to its drawing.









RAY

POINT

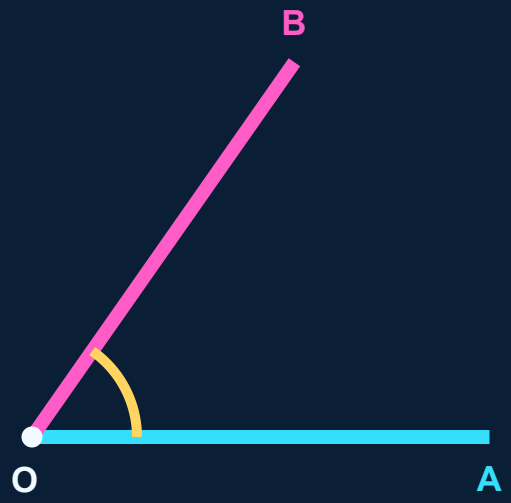
LINE

LINE SEGMENT

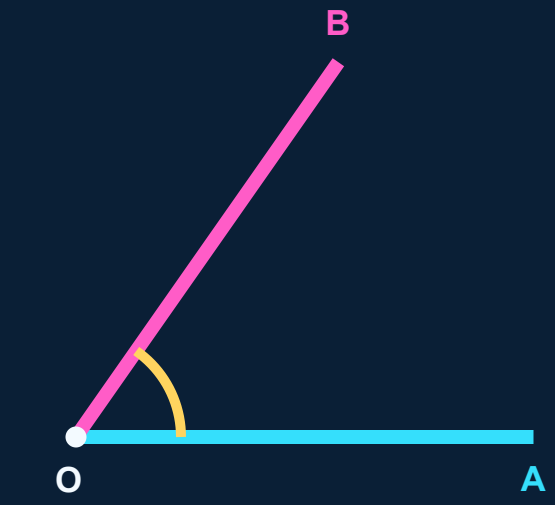
Two rays begin at the same point O.



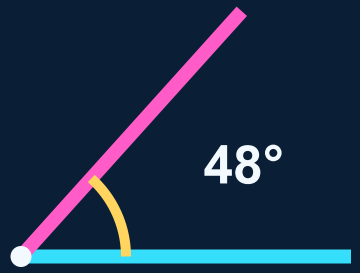
When one ray turns, the two rays form an angle.



O is called the vertex, and OA and OB are the arms.

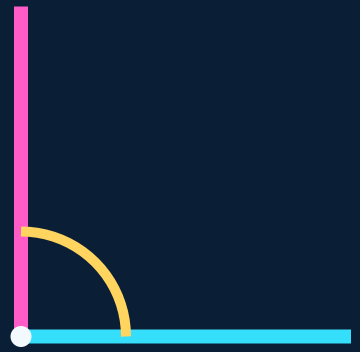


An angle smaller than 90° is acute.



ACUTE

An angle of 90° is a right angle.



RIGHT

90°

An angle between 90° and 180° is obtuse.



OBTUSE

125°

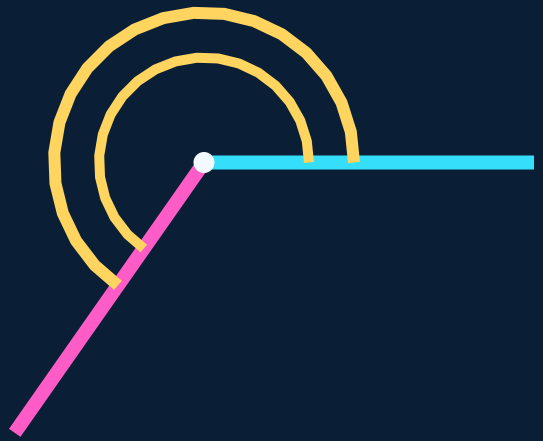
An angle of 180° is straight.



STRAIGHT

180°

An angle between 180° and 360° is reflex.



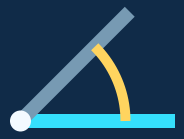
REFLEX

235°

Which angle is obtuse?

A

45°



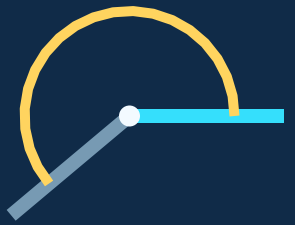
B

125°



C

220°



Which angle is reflex?

A

95°



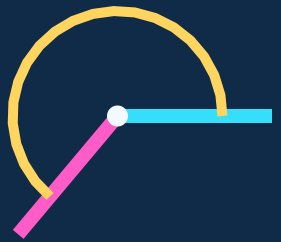
B

180°



C

230°



What is the sum of these angles?



$$35 + 55 = ?$$

Correct! These two angles add to 90°.



35° + 55°

35° + 55° = 90°

Angles whose measures add to 90° are called complementary angles.

COMPLEMENTARY



35° + 55° = 90°

What about these two?



$$125 + 55 = ?$$

Correct! These two angles add to 180°.



125°

55°

125° + 55° = 180°

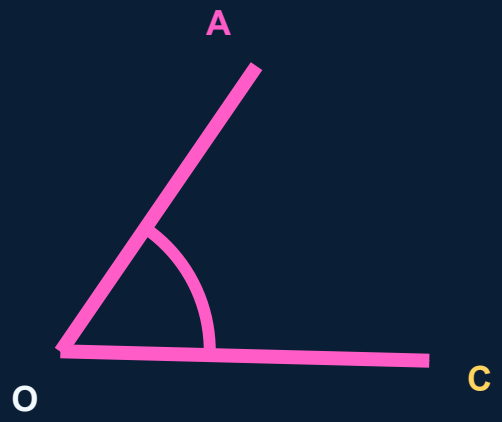
Angles whose measures add to 180° are supplementary.

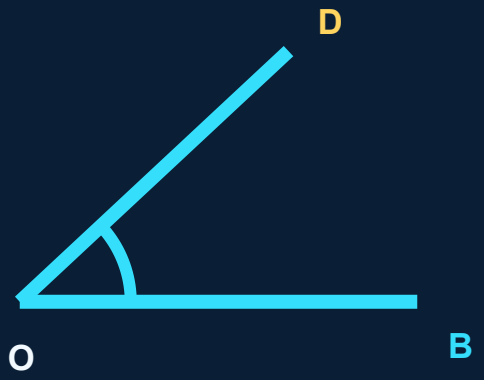
SUPPLEMENTARY



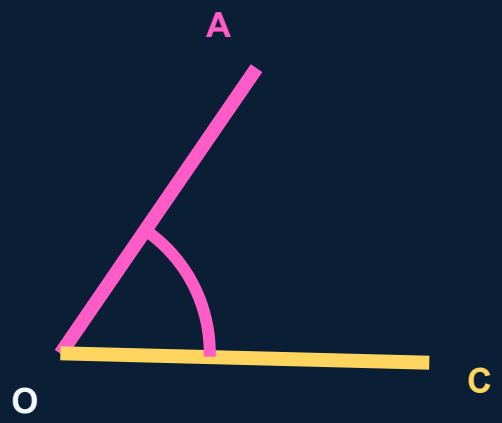
125° + 55° = 180°

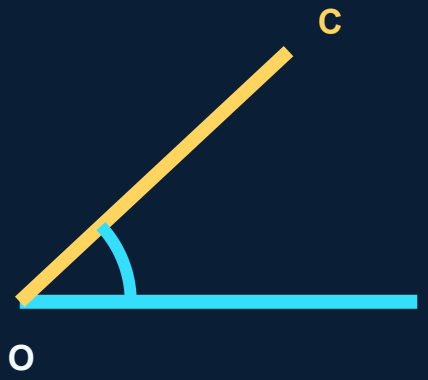
Let’s look at these two angles.



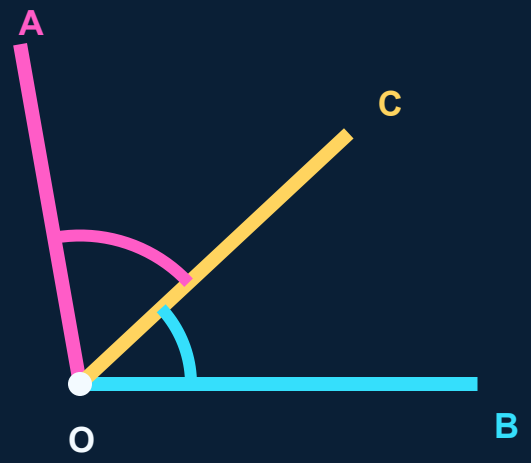


What if they had the same vertex and a common arm?

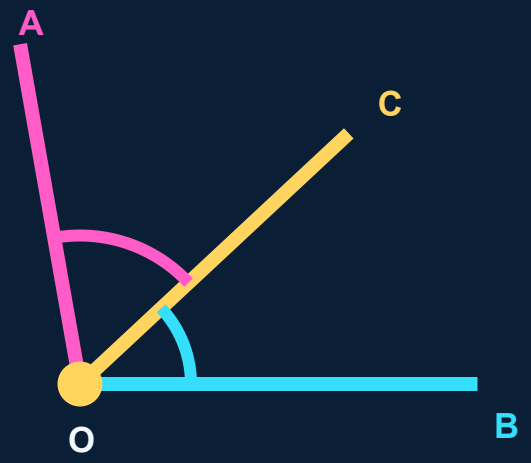




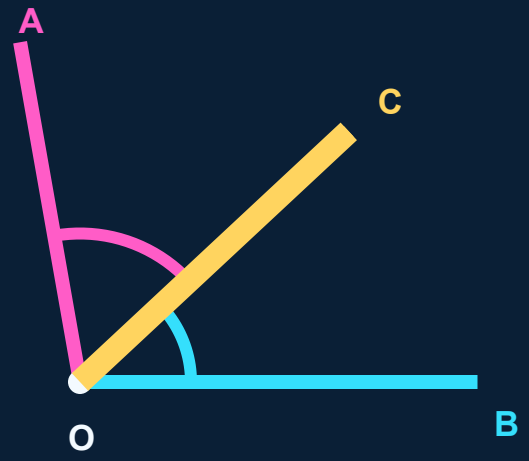
Let’s find out.



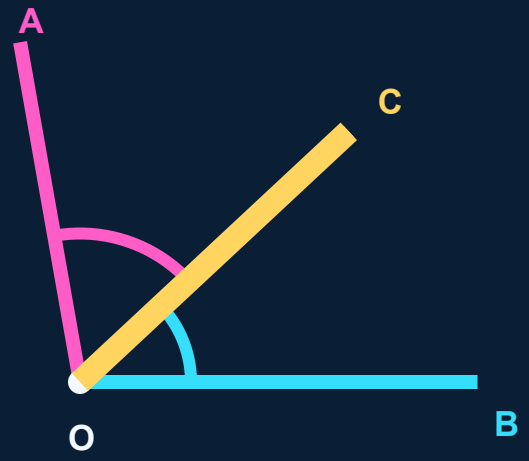
The two angles share vertex O.



They also share arm OC without overlapping.

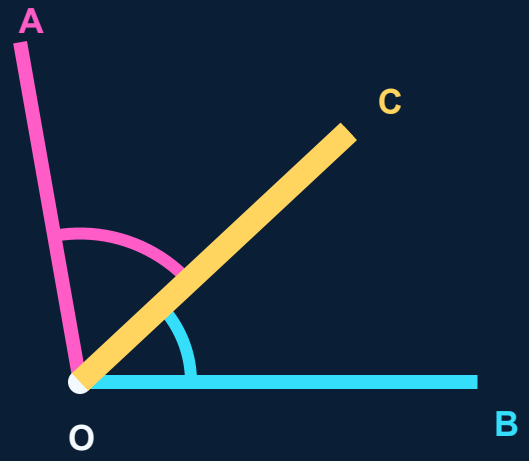


These are adjacent angles.



ADJACENT ANGLES

Two angles which share a common arm and vertex without any overlap are called adjacent angles.

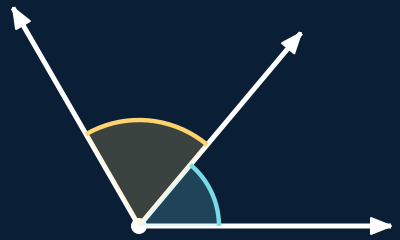


ADJACENT ANGLES

Which of these pairs are adjacent angles?

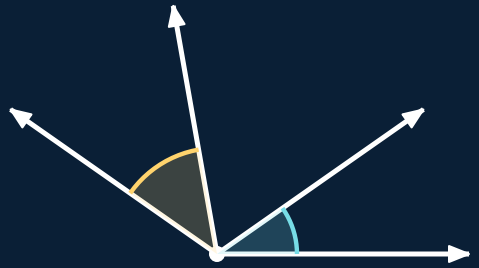
F02

A



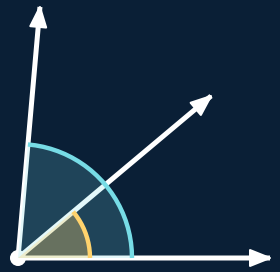
A

B



B

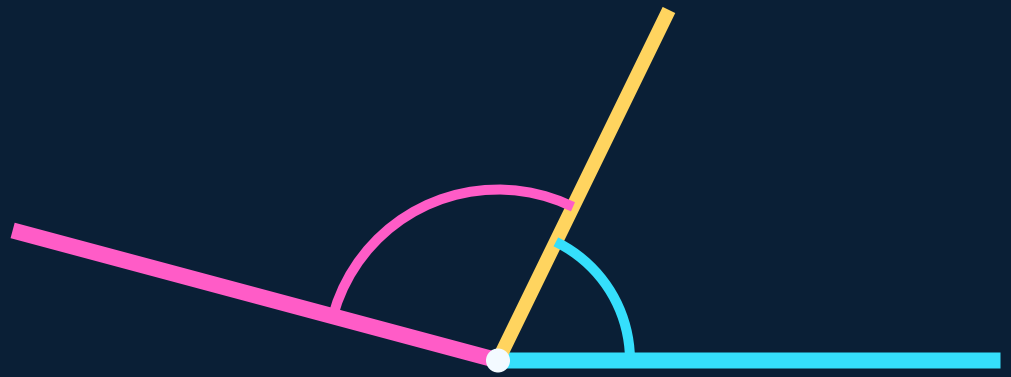
C



C

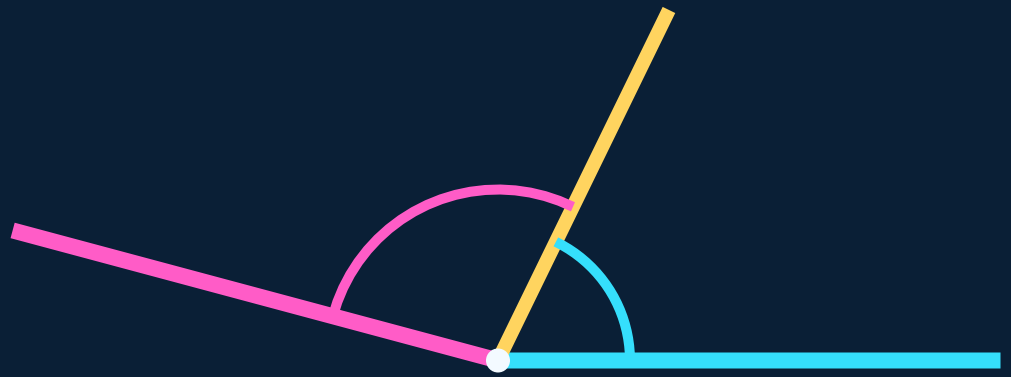
Check

What if their non-common arms lie on a straight line?



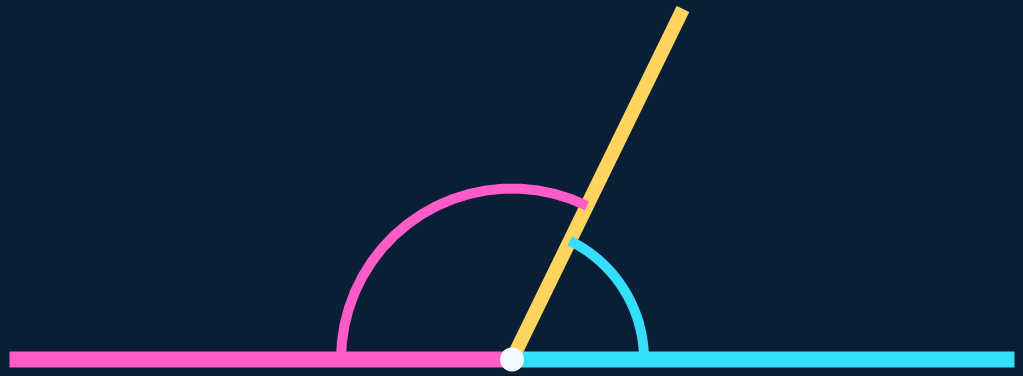
NON-COMMON ARMS

Drag A till OA and OB lie on a straight line.



NON-COMMON ARMS

The outer arms now make one straight line.

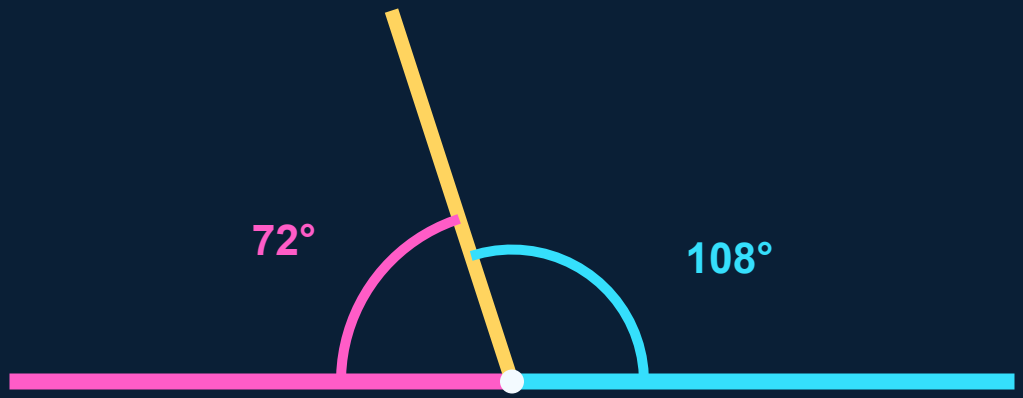


STRAIGHT LINE

Move the common arm and watch both measures.



One angle grows as the other shrinks.



What stays the same when the common arm moves?

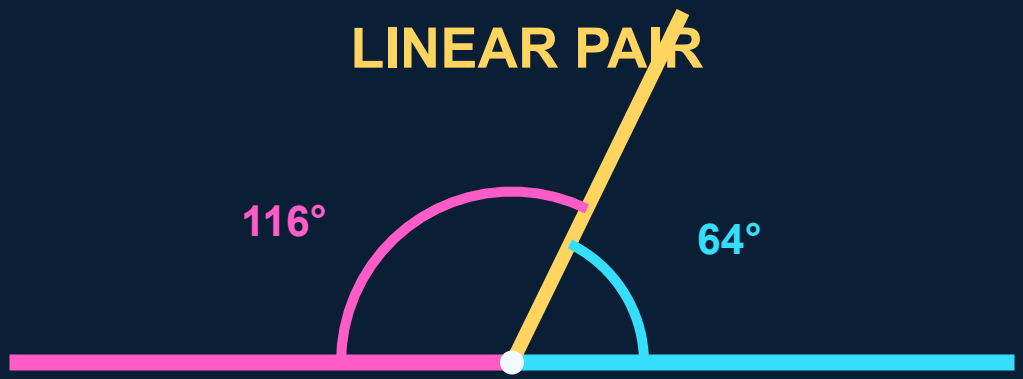


The two adjacent angles always add to 180°.

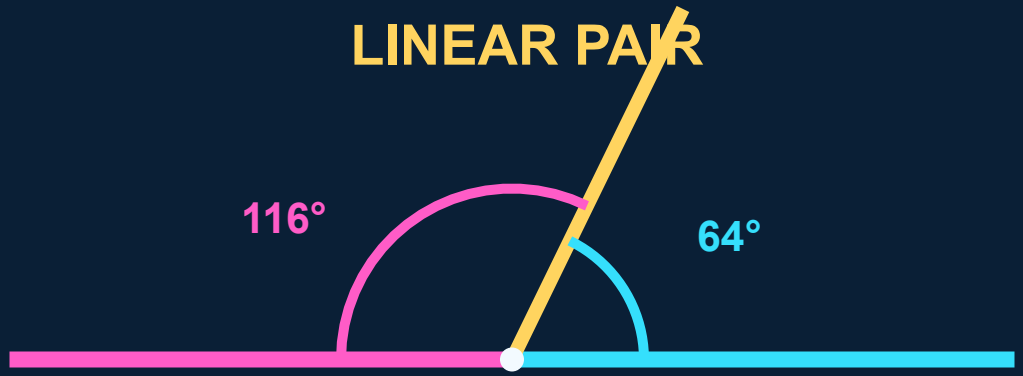


180°

A ray standing on a line forms a linear pair.



The angles in a linear pair add to 180°.



180°

Which diagram shows a linear pair?

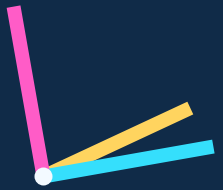
A



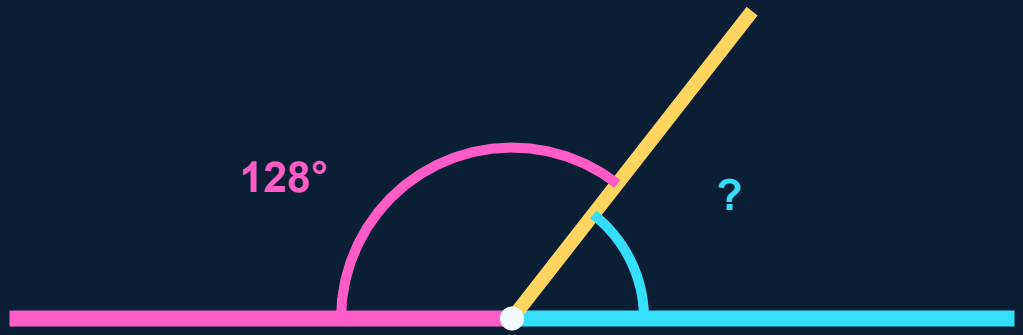
B



C

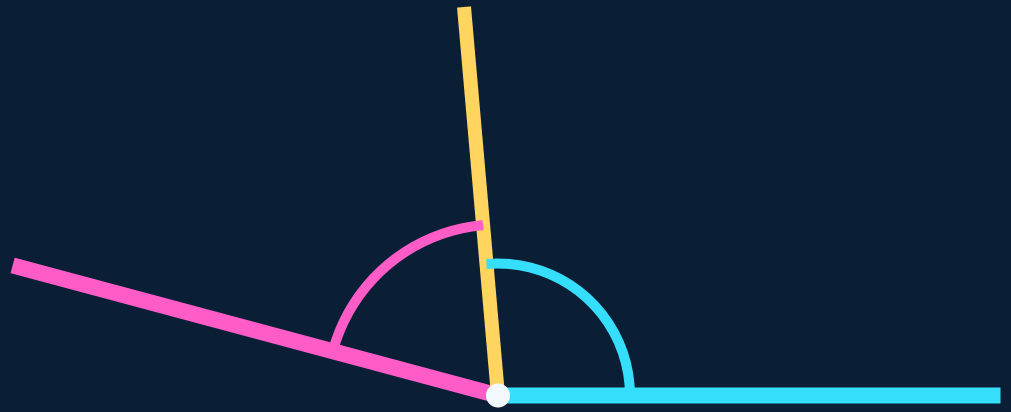


One angle is 128°; what is the other angle?



$$128° + ? = 180°$$

Now reverse the relationship.



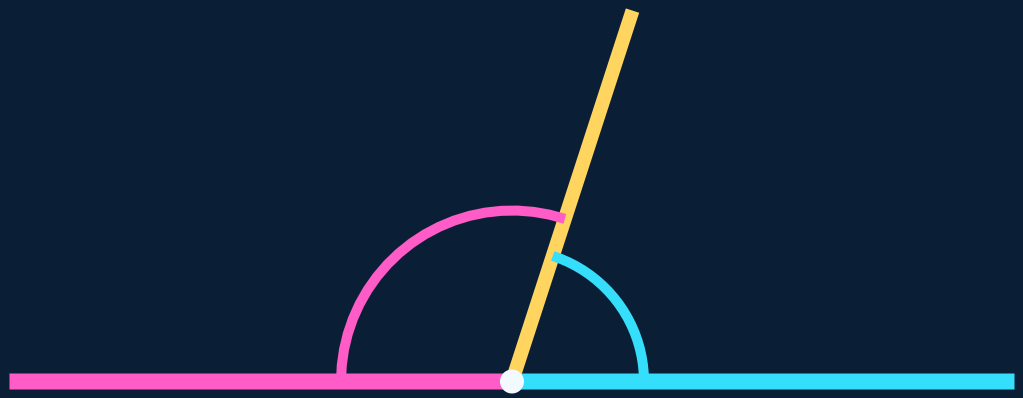
?

Make the adjacent angles add to 180°.



TARGET 180°

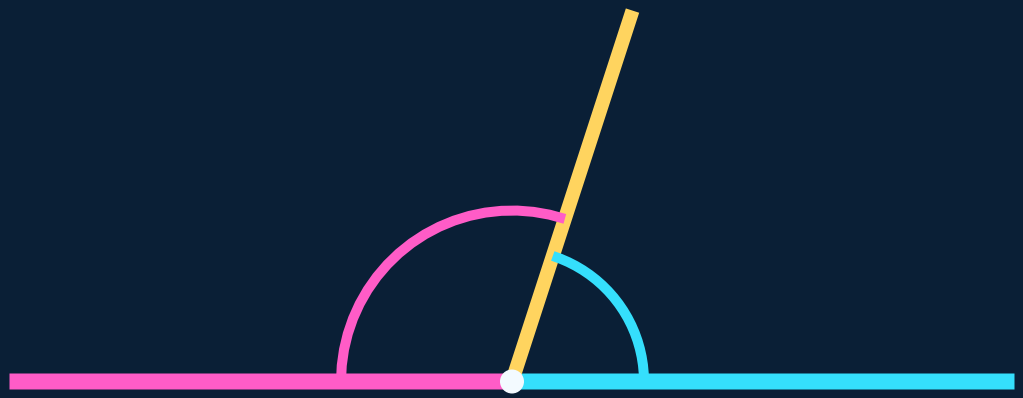
What happens to their non-common arms?



180°

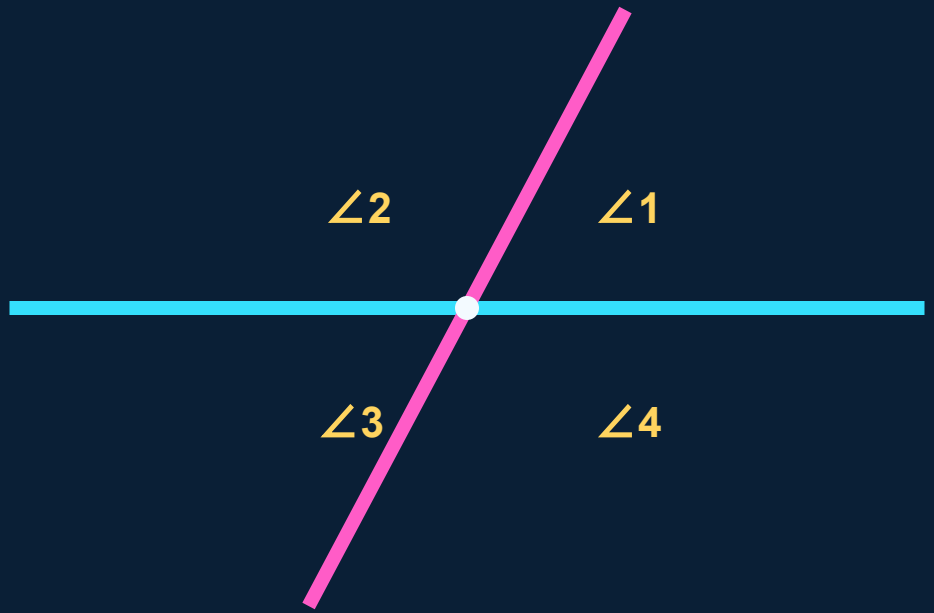
STRAIGHT LINE

If adjacent angles add to 180°, their non-common arms form a line.

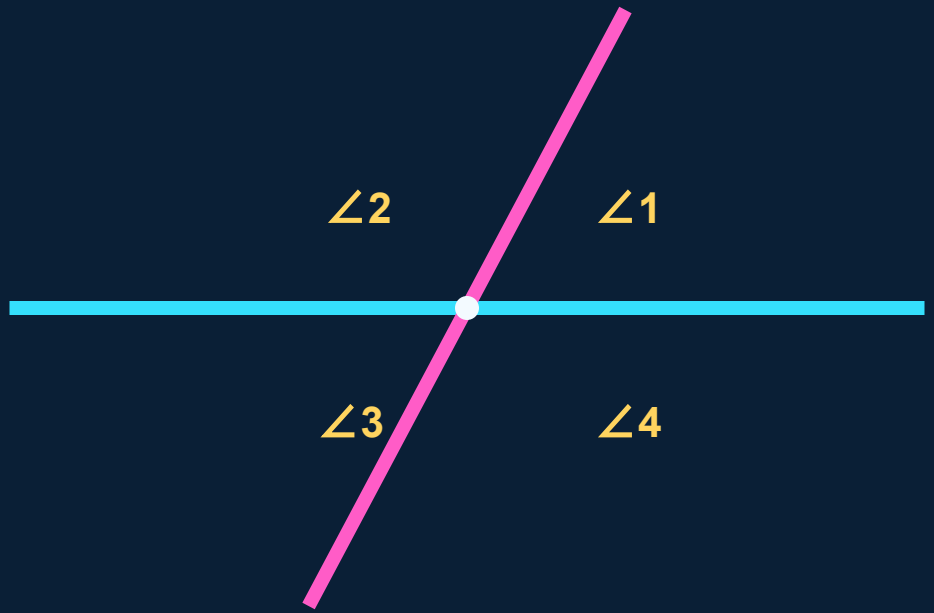


180°

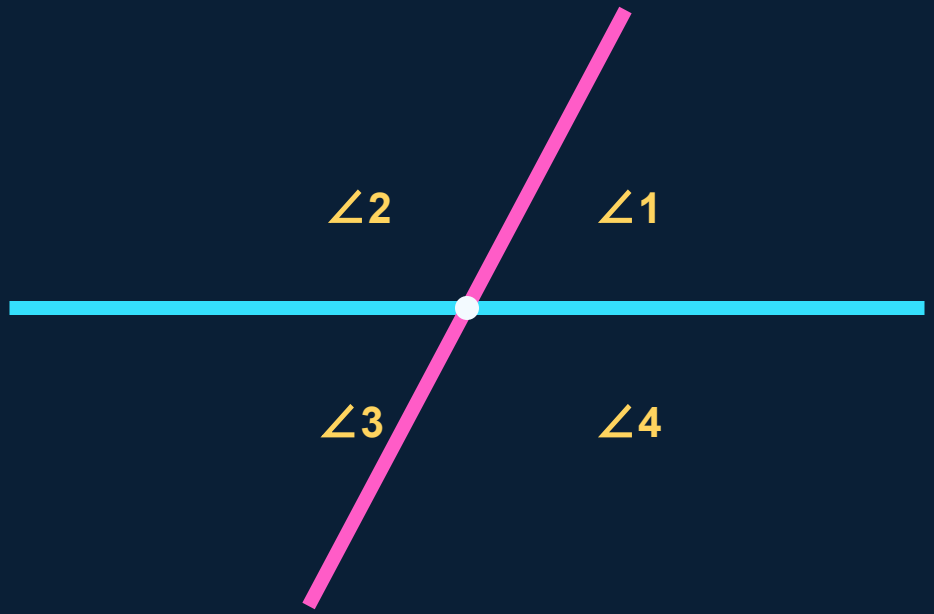
What happens when a whole line crosses another line?



How many angles can you see?

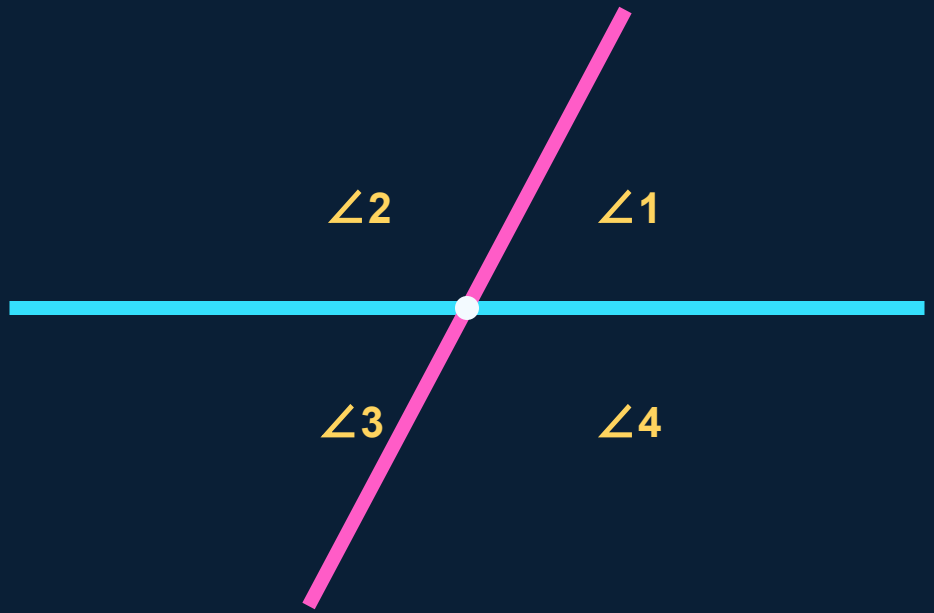


Four angles are formed.

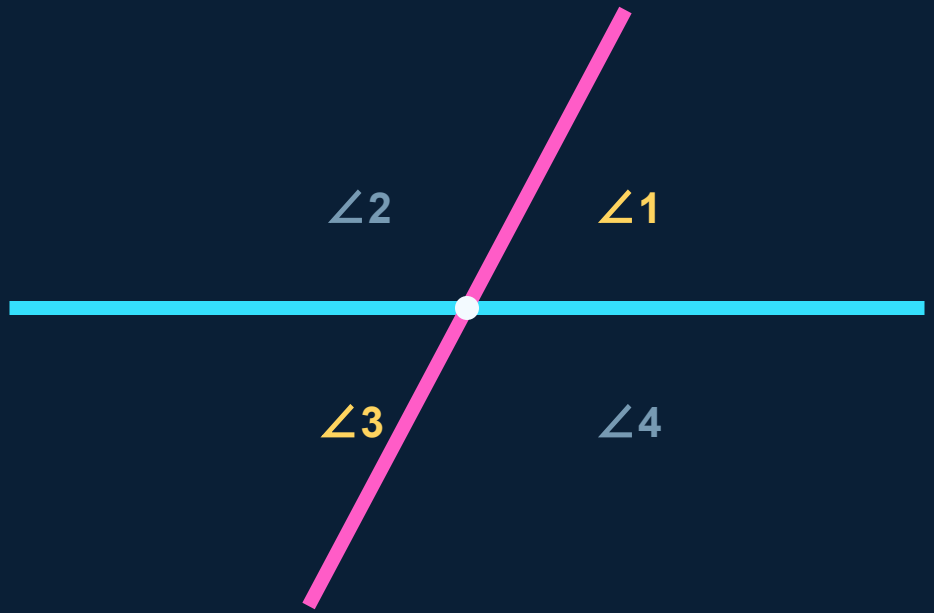


4 angles

Which angle is opposite ∠1?

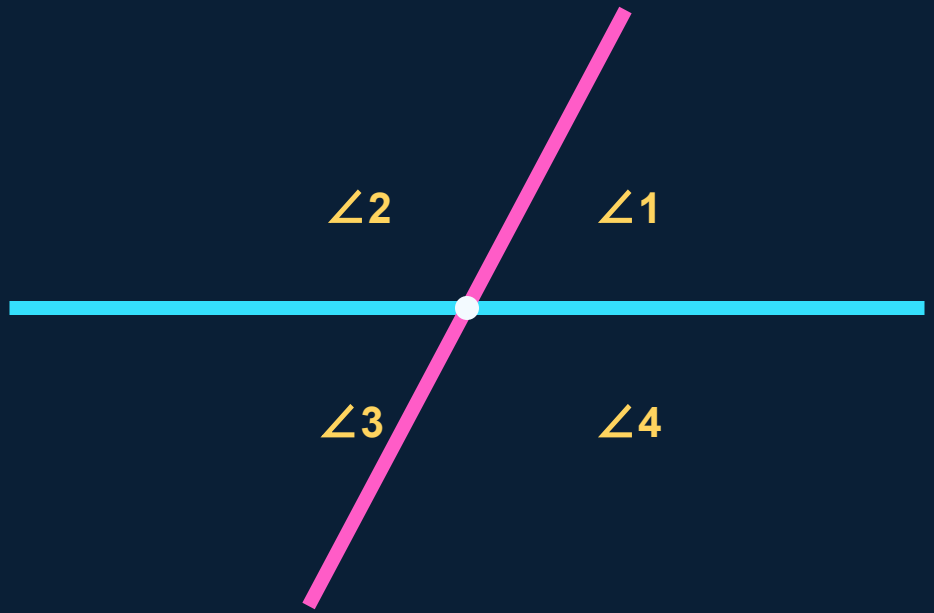


∠3 lies opposite ∠1.

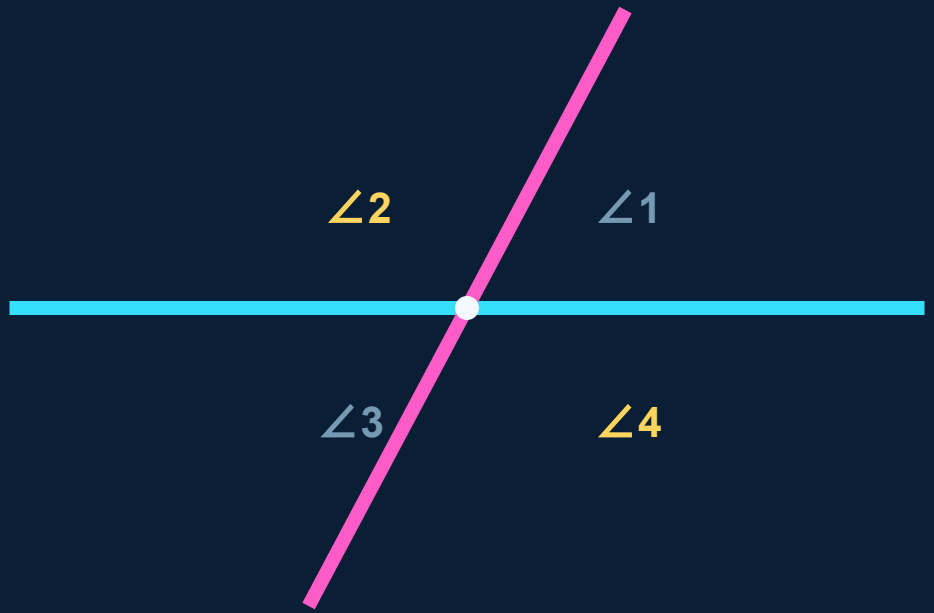


∠1 ∠3

Which angle is opposite ∠2?

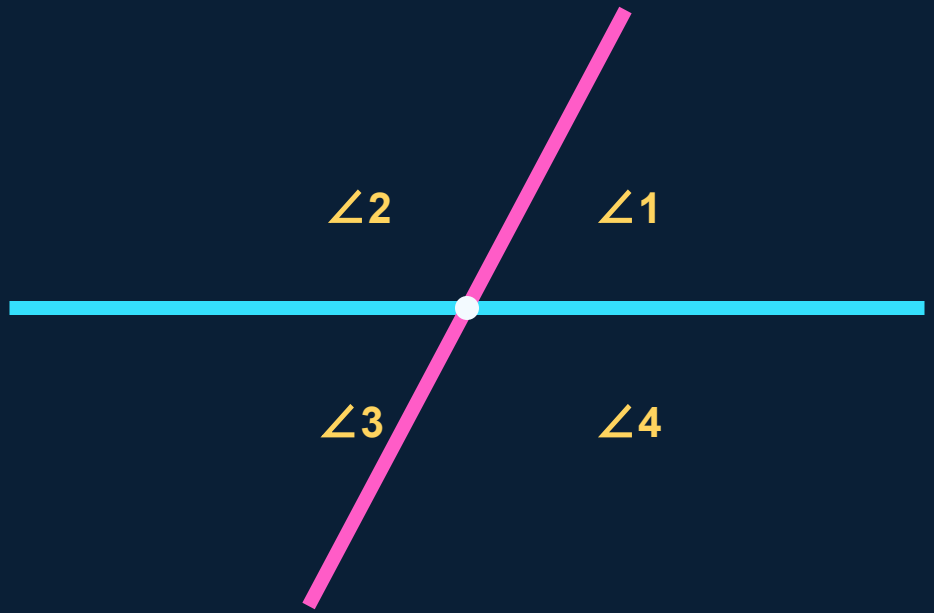


∠4 lies opposite ∠2.



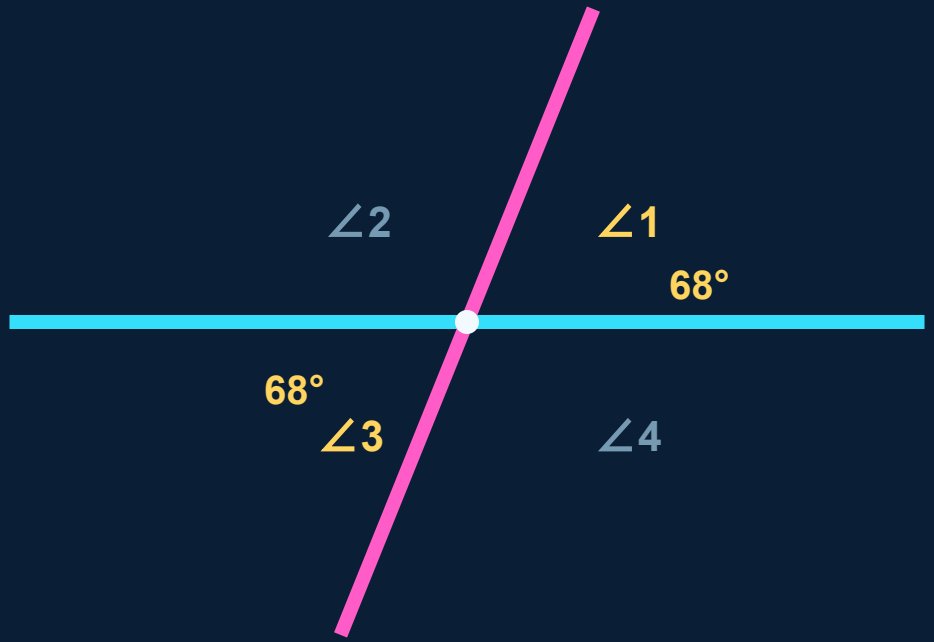
∠2 ∠4

These pairs are called vertically opposite angles.

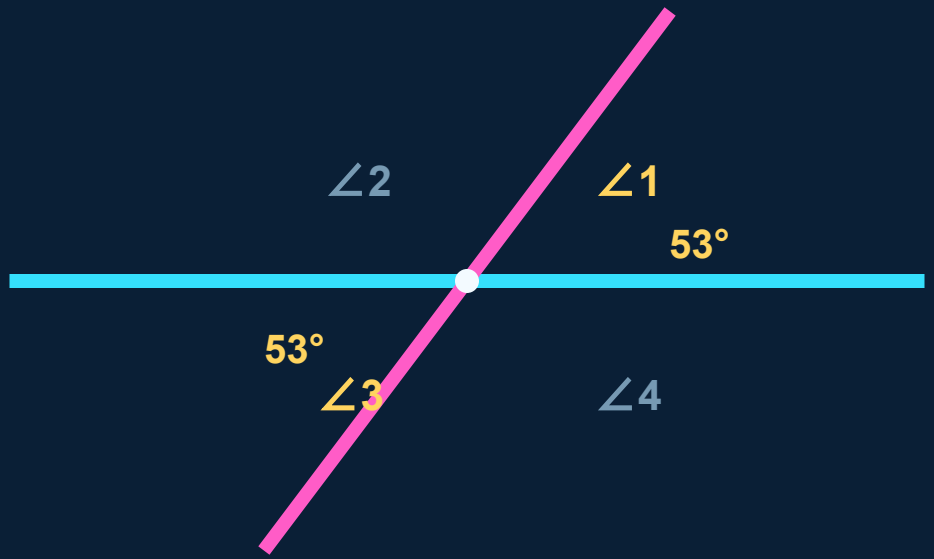


VOA

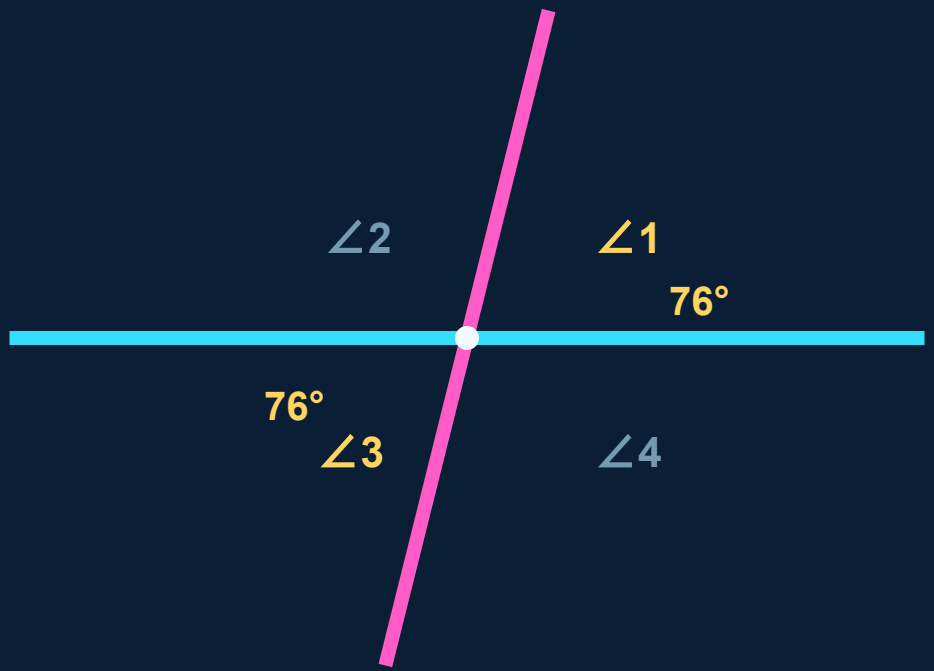
Rotate one line and watch ∠1 and ∠3.



Their measures change together.

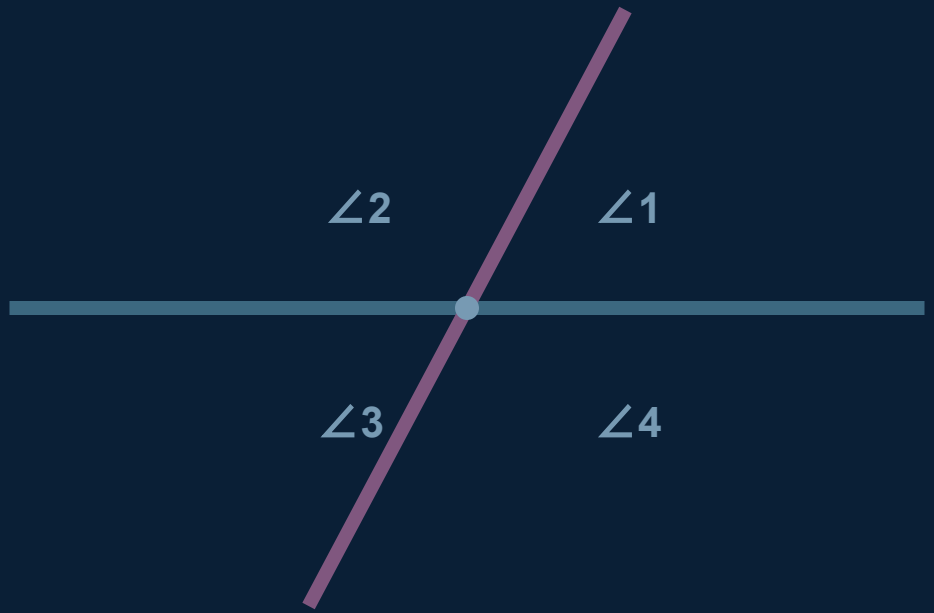


What stays the same about ∠1 and ∠3?



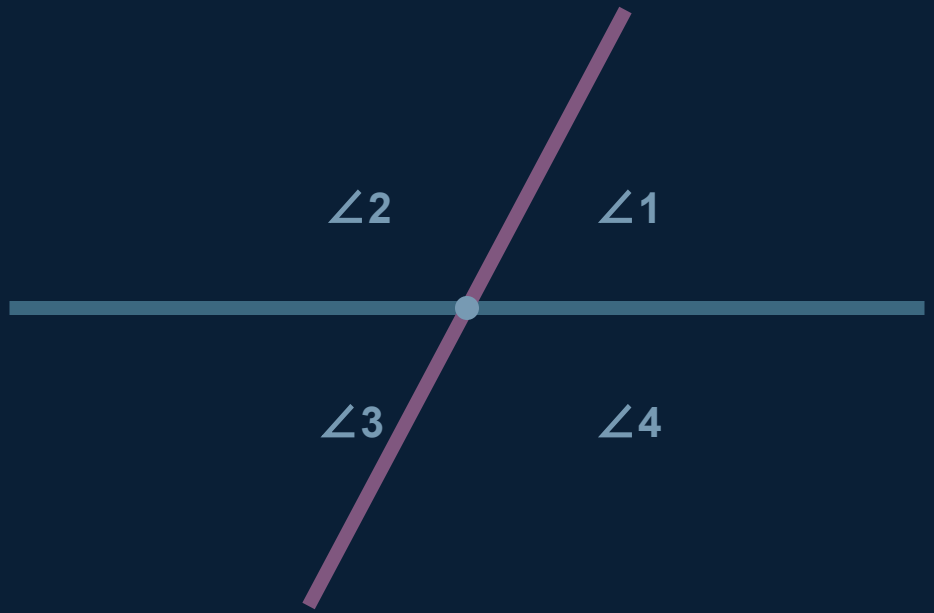
∠1 = ∠3

Measuring suggests that opposite angles are equal.



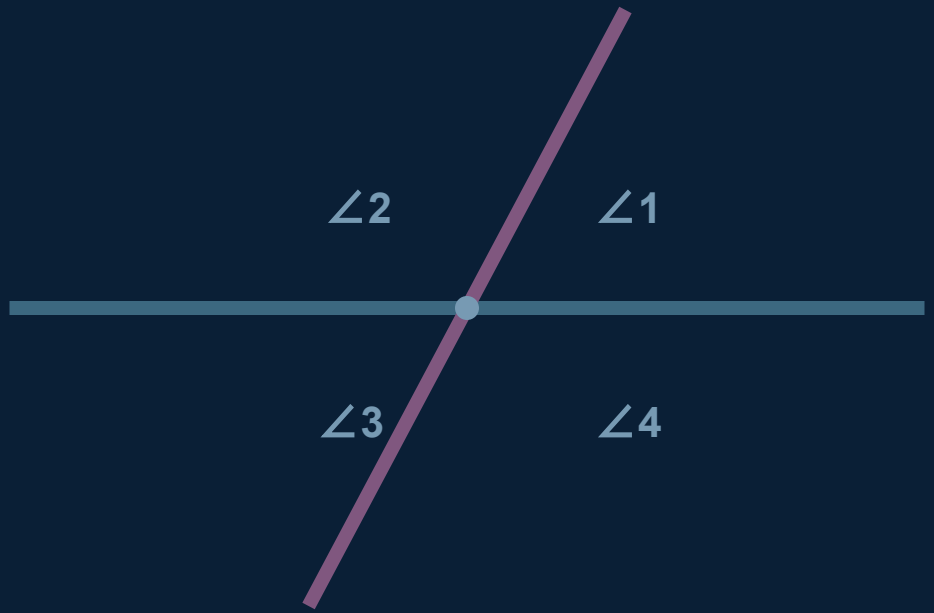
$\angle 1 = \angle 3$

Can we prove that ∠1 = ∠3?



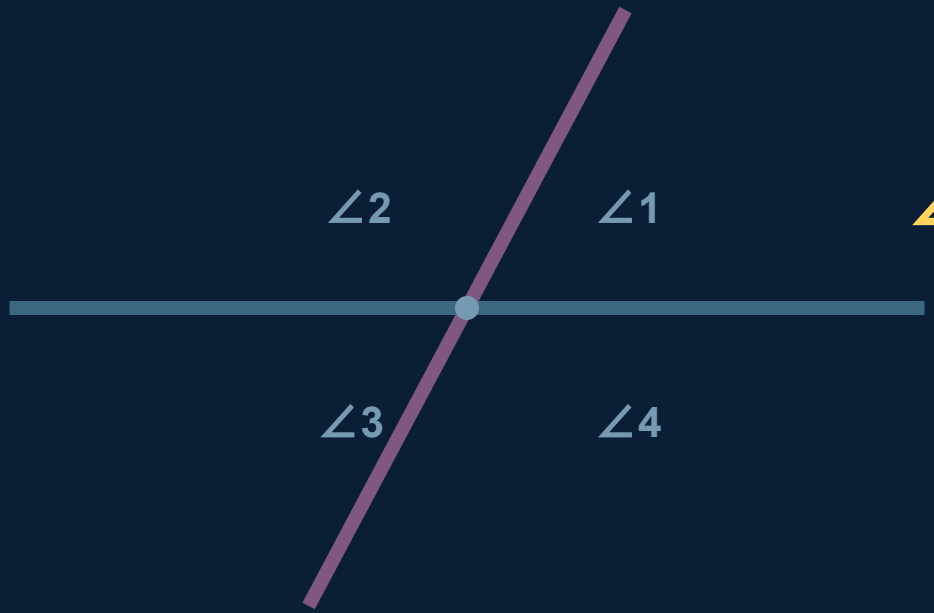
∠1 = ∠3

∠1 and ∠2 form a linear pair.



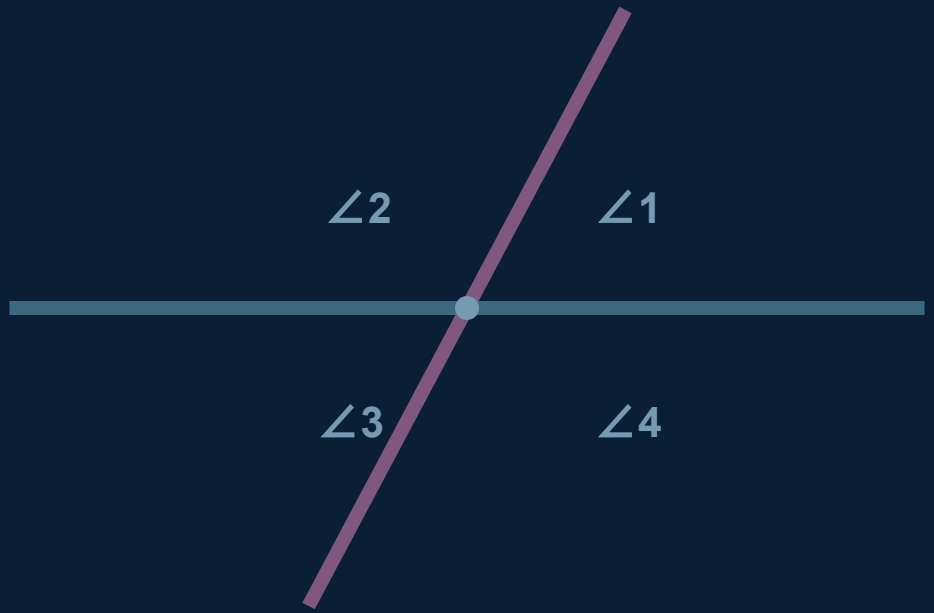
∠1 + ∠2

Therefore, ∠1 + ∠2 = 180°.



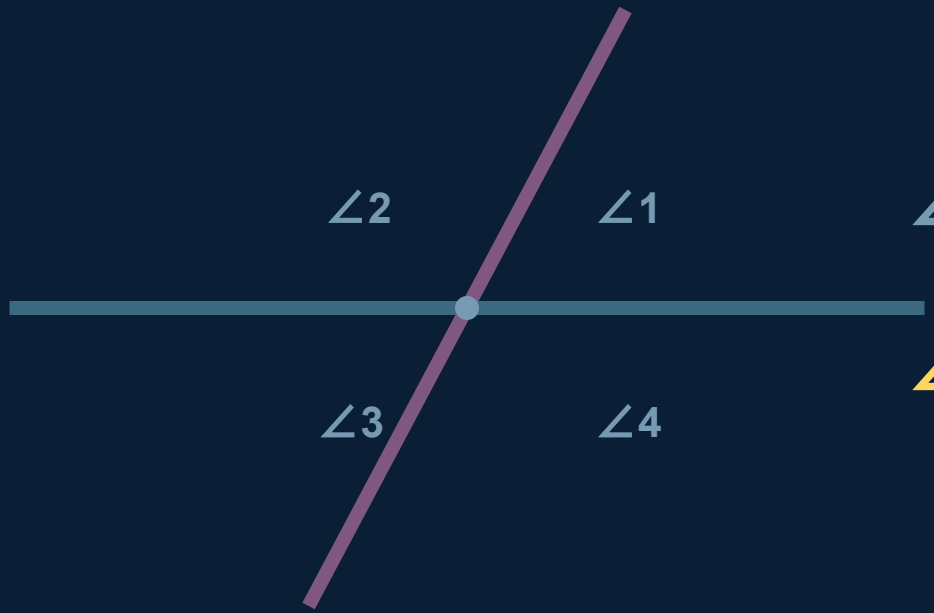
∠1 + ∠2 = 180°

∠2 and ∠3 also form a linear pair.



∠2 + ∠3

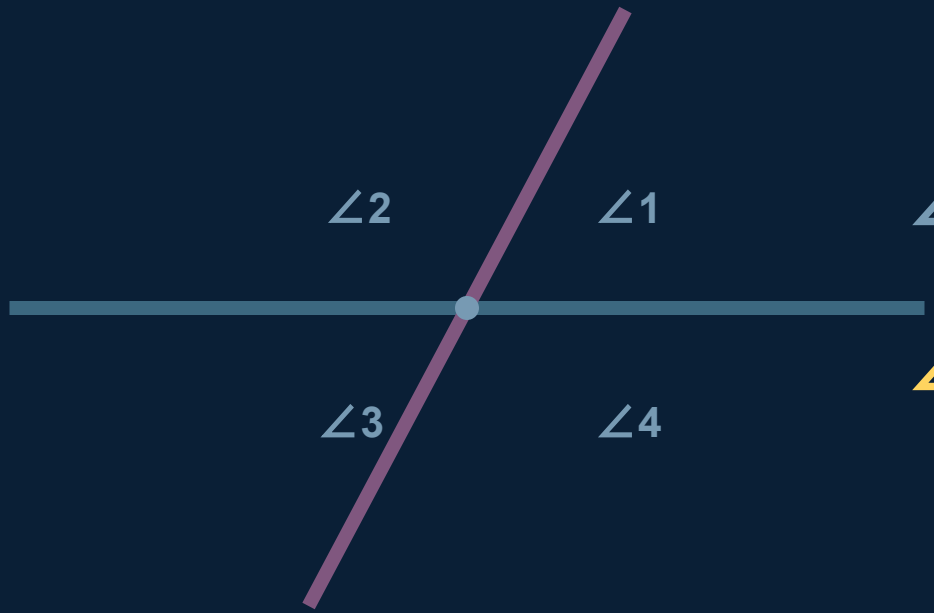
Therefore, $\angle 2 + \angle 3 = 180°$.



$\angle 1 + \angle 2 = 180°$

$\angle 2 + \angle 3 = 180°$

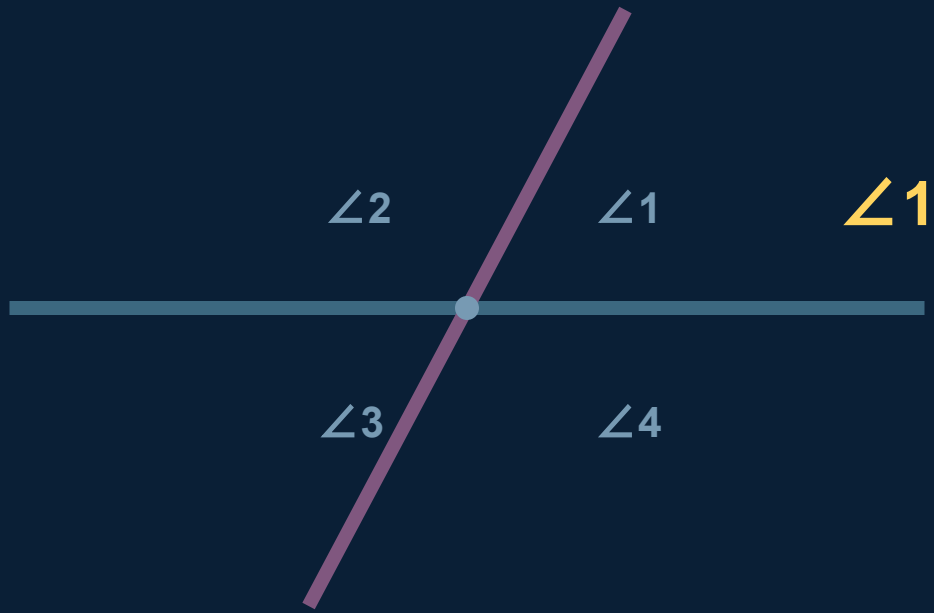
Both expressions are equal to 180°.



$\angle 1 + \angle 2 = 180°$

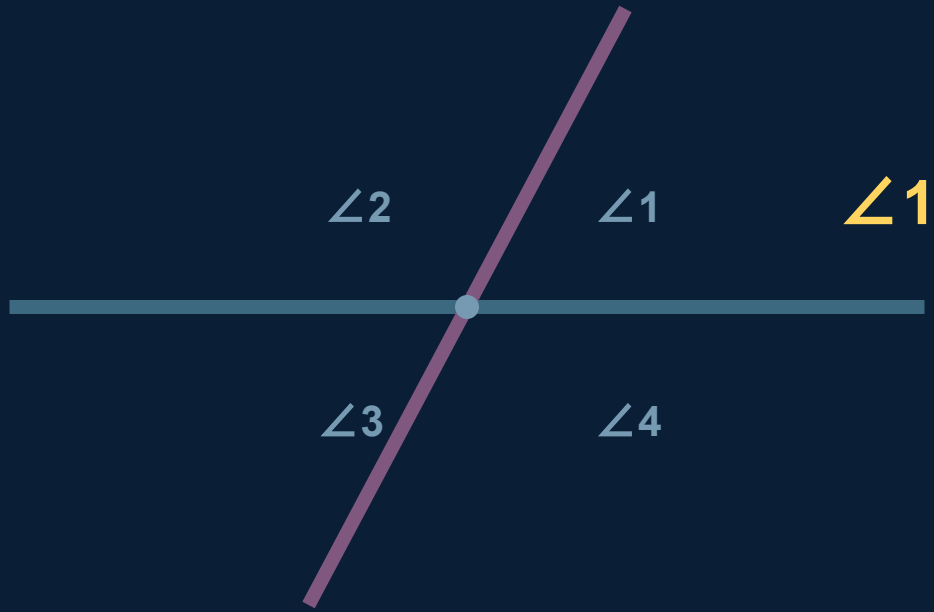
$\angle 2 + \angle 3 = 180°$

So ∠1 + ∠2 = ∠2 + ∠3.



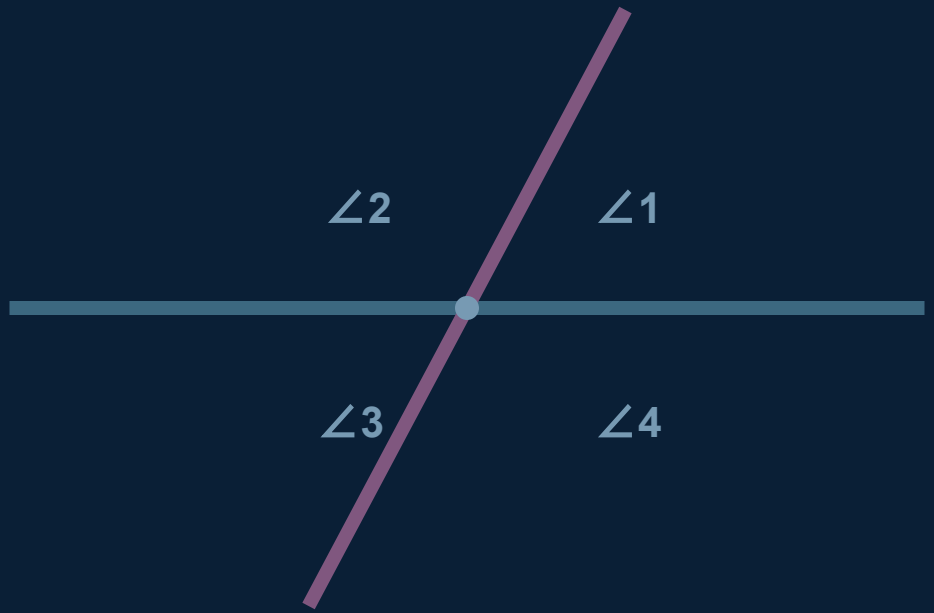
∠1 + ∠2 = ∠2 + ∠3

Subtract the same ∠2 from both sides.



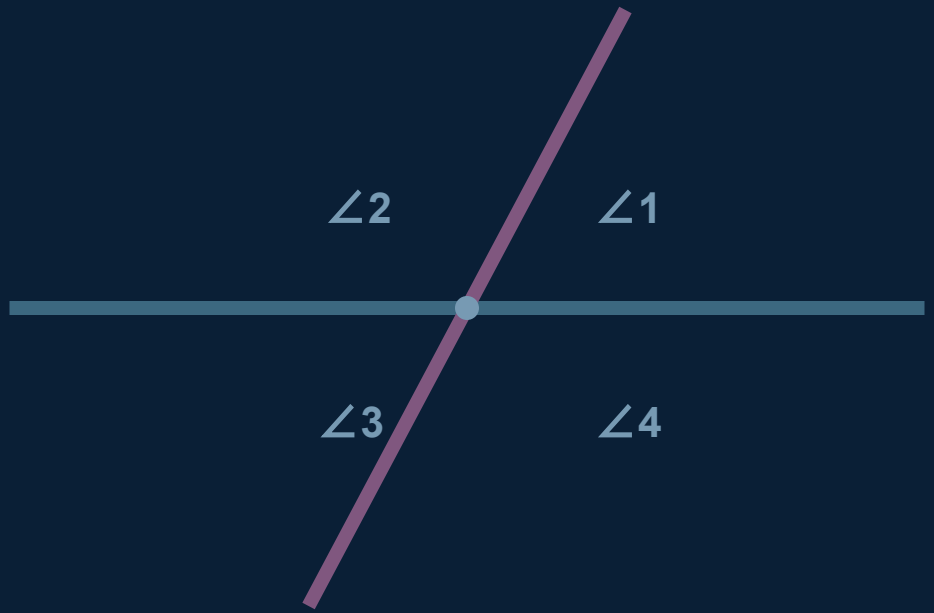
$$\angle 1 + \angle 2 = \angle 2 + \angle 3$$

We are left with $\angle 1 = \angle 3$.



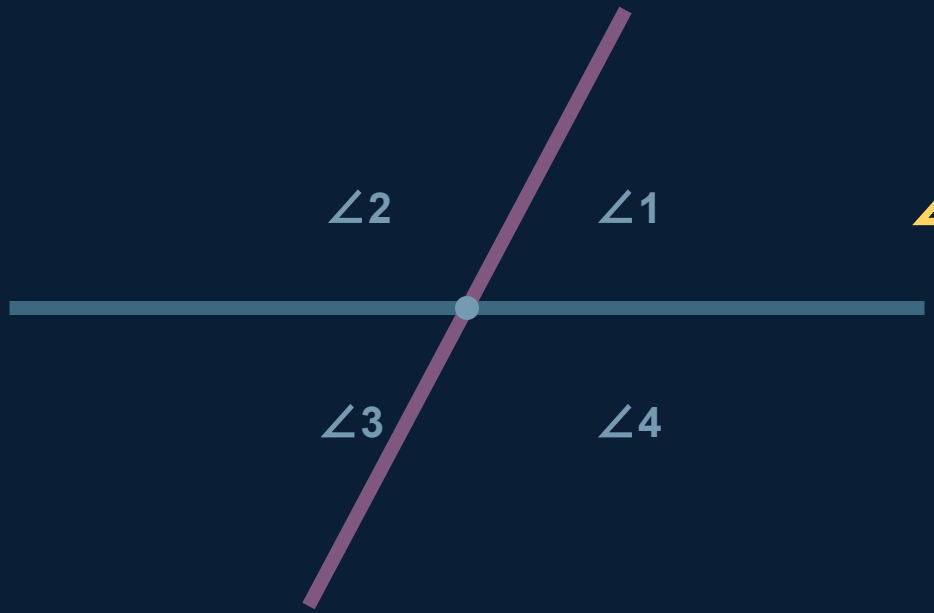
$\angle 1 = \angle 3$

Can you prove that ∠2 = ∠4?



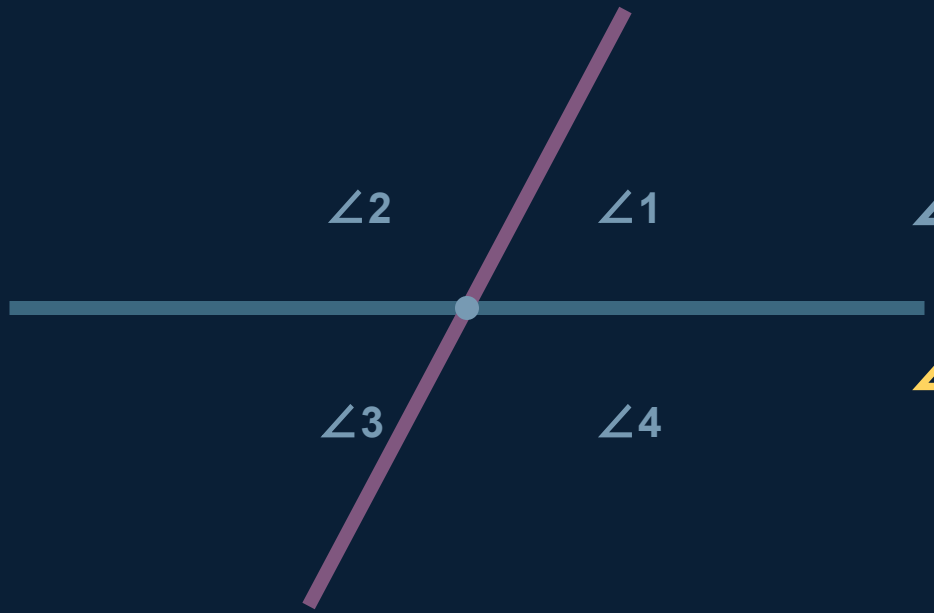
∠2 = ∠4

$\angle 1 + \angle 2 = 180°$.



$\angle 1 + \angle 2 = 180°$

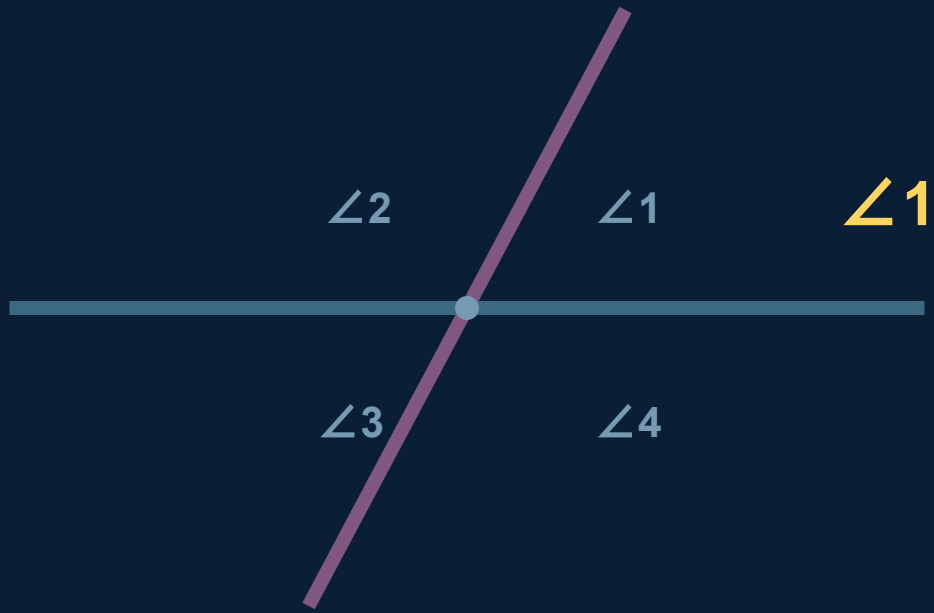
$\angle 1 + \angle 4 = 180°$.



$\angle 1 + \angle 2 = 180°$

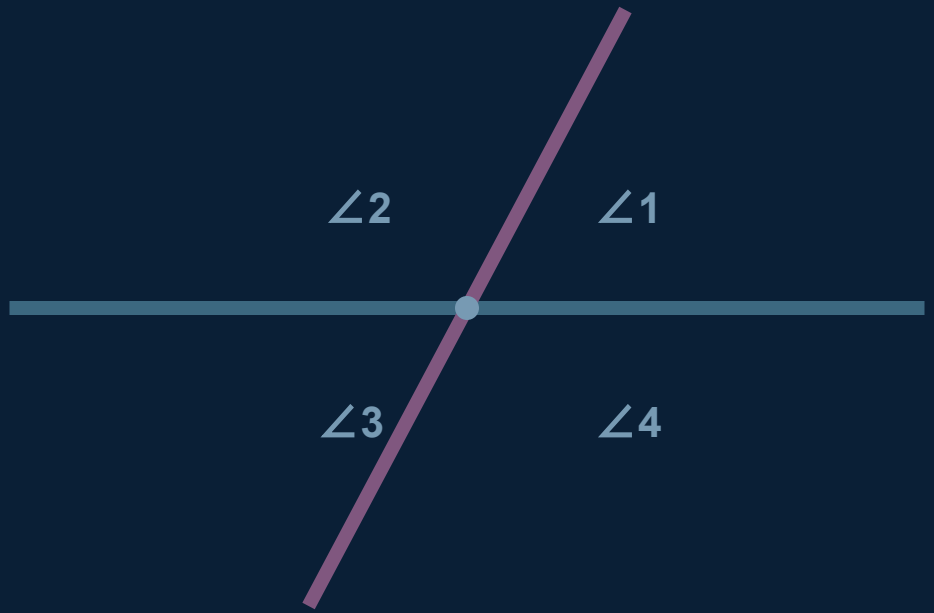
$\angle 1 + \angle 4 = 180°$

Subtract ∠1 from both sides.



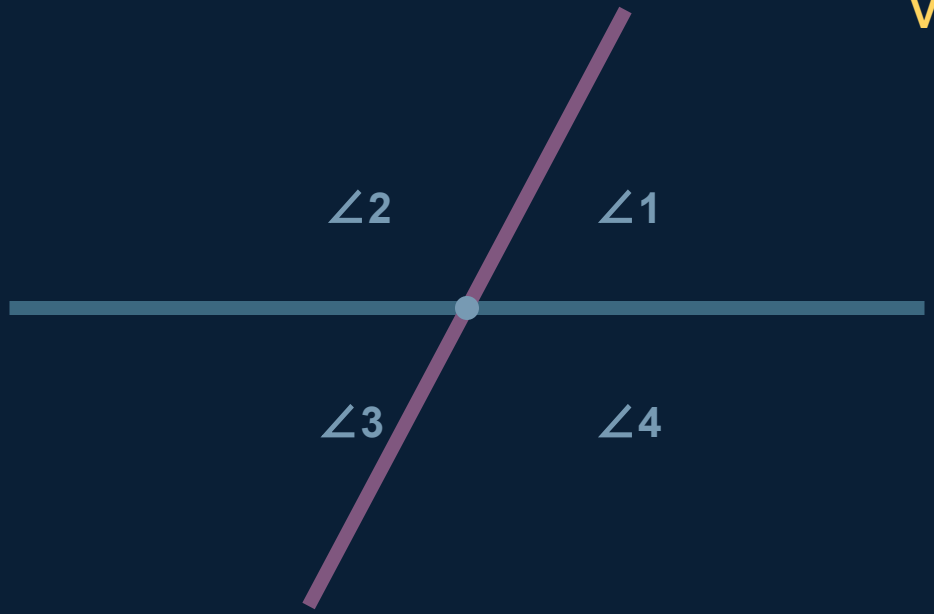
$$\angle 1 + \angle 2 = \angle 1 + \angle 4$$

Therefore, ∠2 = ∠4.



∠2 = ∠4

When two lines intersect, vertically opposite angles are equal.

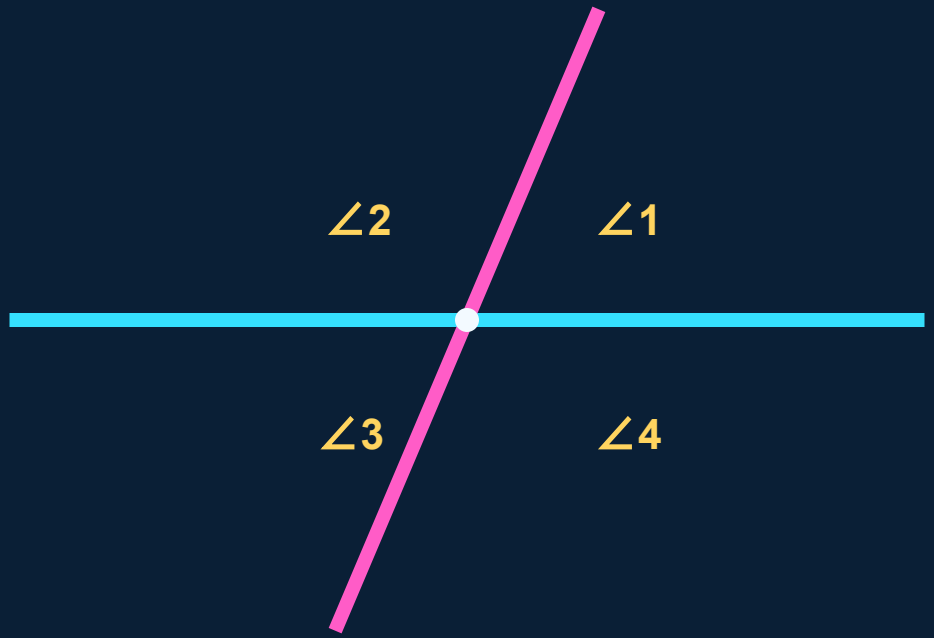


VERTICALLY OPPOSITE ANGLES

$\angle 1 = \angle 3$

$\angle 2 = \angle 4$

If ∠1 is 67°, what is ∠3?

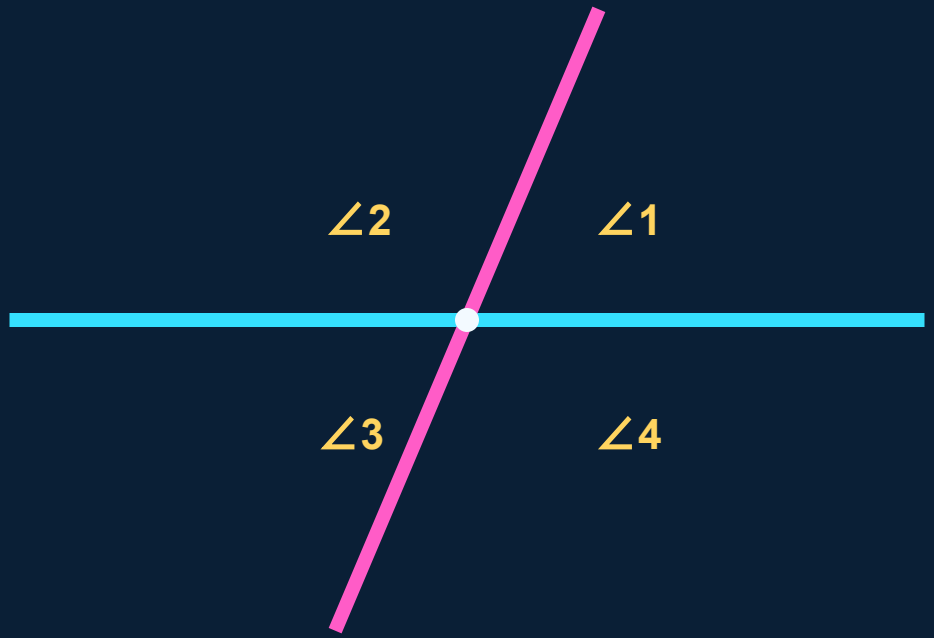


∠1 = 67°

∠3

?

What is ∠2 if it forms a linear pair with 67°?



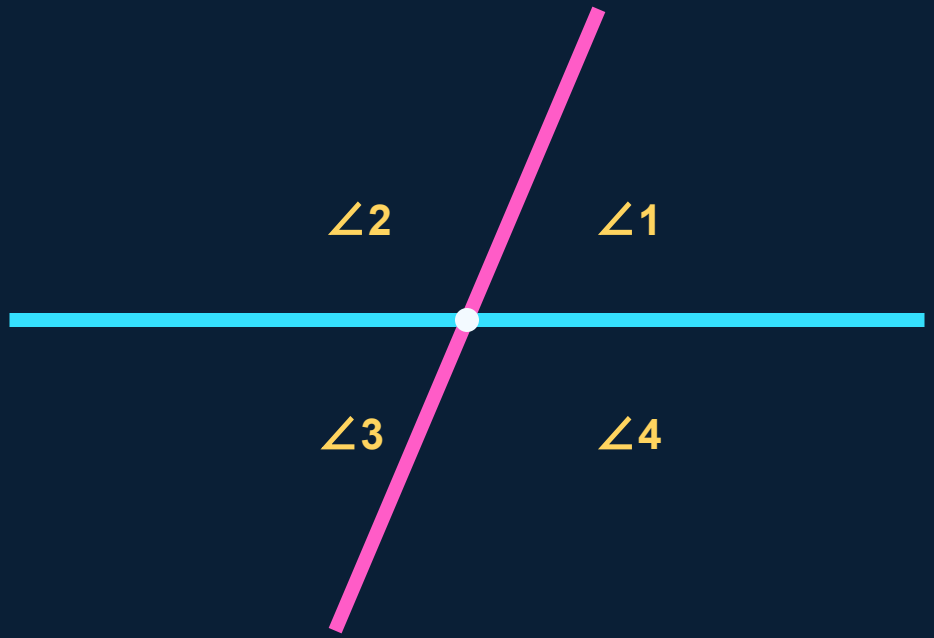
∠1 = 67°

∠3 = 67°

∠2

?

What is ∠4 now?



∠1 = 67°

∠3 = 67°

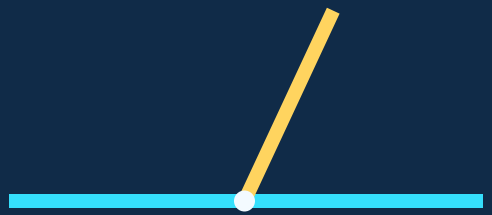
∠2 = 113°

∠4

?

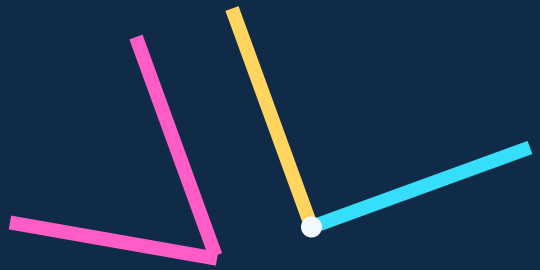
Both pairs total 180°; which pair can use the converse?

PAIR A



180°

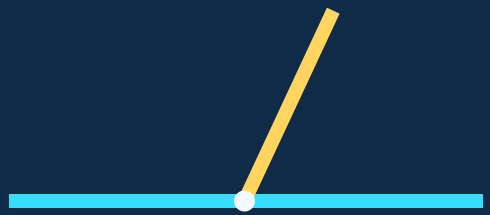
PAIR B



180°

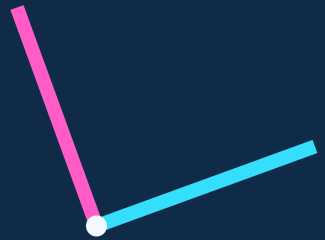
Why does only Pair A meet the converse conditions?

PAIR A



180°

PAIR B



180°

ADJACENT

The calibration system is ready for the asteroid mission.

